

PROJECT PRESENTATION

Department of Computer Science and Engineering

KLH Bachupally Campus, Hyderabad

Team __ - Section __

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B.Tech CSE, A.Y. 2026-27

Guide Name

Mr. Shaik Asif

Project Title :

Cricket Player Performance Prediction Using Ridge, Lasso and Elastic Net Regression

INTRODUCTION

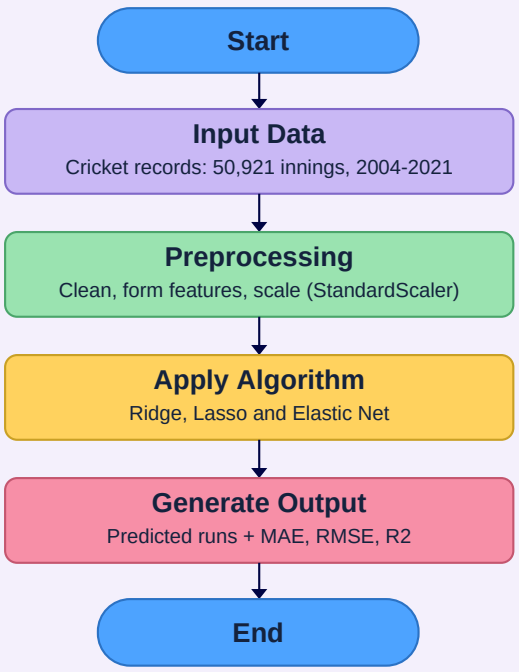
- **Overview:** A machine-learning regression system that predicts the **runs a batter will score in an innings** from previous match records (50,921 innings, 1,571 players).
- **Problem:** Weighing balls faced, boundaries, batting order and recent form by eye is subjective and does not scale.
- **Objectives:** Train and compare **Ridge, Lasso** and **Elastic Net**; evaluate with MAE, MSE, RMSE and R^2.
- **Scope & significance:** Supports team selection, fantasy scoring and workload planning, and shows which factors carry real signal.

RESEARCH GAP

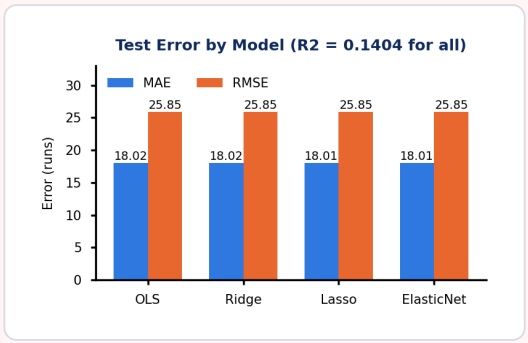
- **Limitation:** A career average cannot notice a recent slump, and one statistic alone does not describe a batter.
- **Unexplored:** How strongly correlated predictors (r up to 0.94) affect coefficients, and what L1 / L2 penalties add.
- **Need:** Leakage-free form features and a fair comparison on one identical pipeline.
- **Our work:** Builds career and recent-form features, scales them, and compares three regularised models on MAE, MSE, RMSE and R^2.

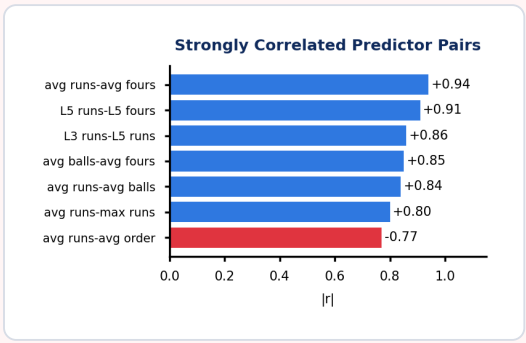
CURRENT RESEARCH → OUR SOLUTION

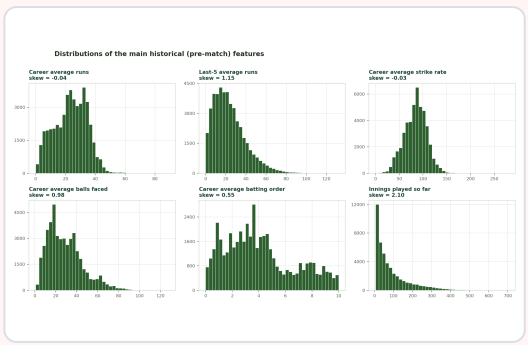
ALGORITHM / WORKFLOW

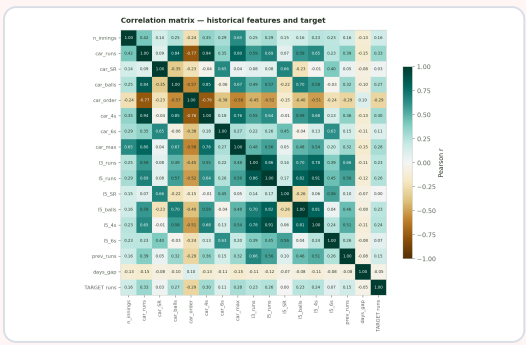


GRAPHS









Dataset: 50,921 batting innings by 1,571 players, 2004-2021 | 10,185 unseen test innings.



RESULTS - FRONT END

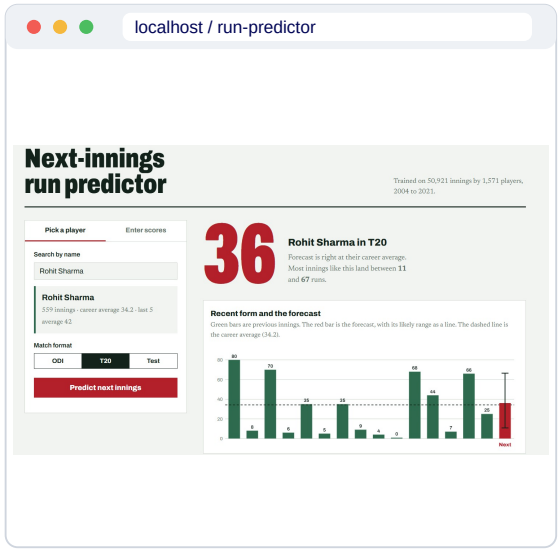
Next-innings Run Predictor: Cricket Player Performance Web App

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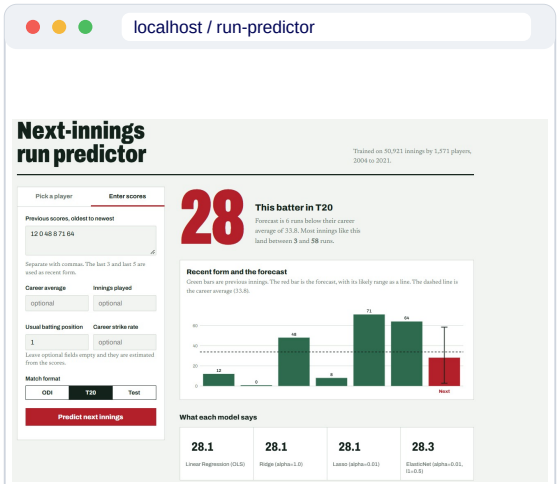
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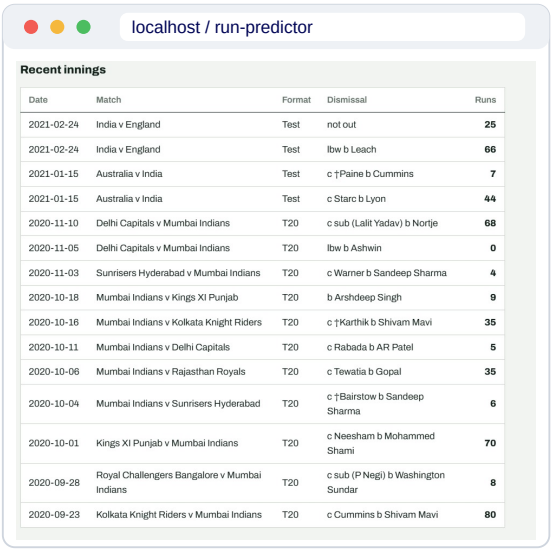
1. Overview (Pick a Player)



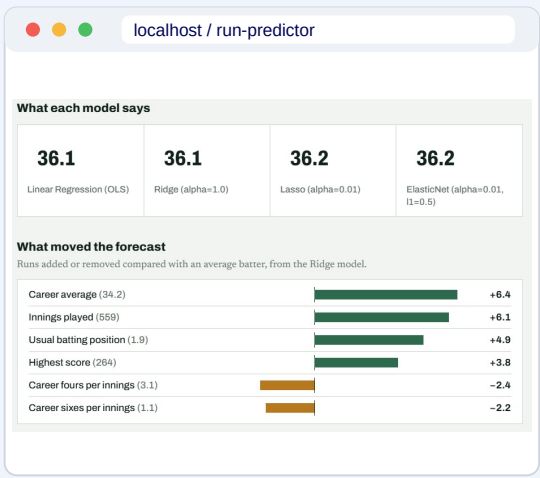
2. Predict Page (Enter Scores)



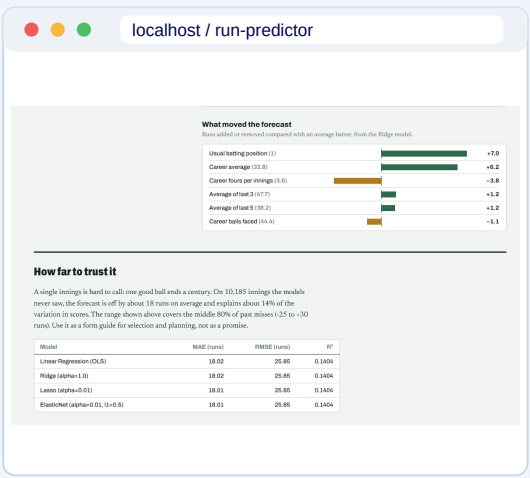
3. Recent Innings



4. ADDITIONAL SCREENS



Model comparison and what moved the forecast



How far to trust it: MAE, RMSE, R2

KEY FRONT END FEATURES

- Pick a player by name or enter previous scores
- Choose match format: ODI, T20 or Test
- Forecast of next-innings runs with a likely range
- Recent form chart: past innings vs the forecast
- Four models side by side: OLS, Ridge, Lasso, ElasticNet
- "What moved the forecast" from the Ridge model

Key Findings

On **10,185 unseen innings**, all four models are off by about **18 runs** on average (MAE 18.01-18.02, RMSE 25.85) and explain about **14%** of the variation in scores (R^2 0.1404). Regularisation barely changes the result, so the forecast is best used as a **form guide** for selection and planning, with the shown range covering the middle 80% of past misses (-25 to +30 runs).